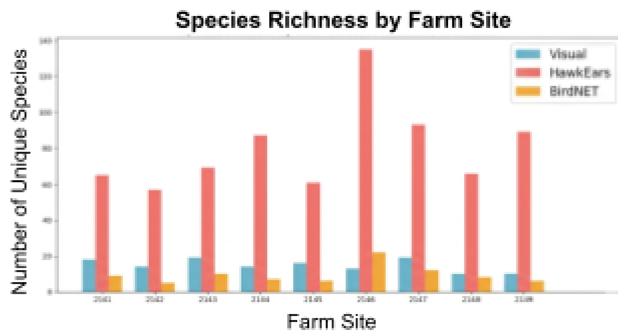
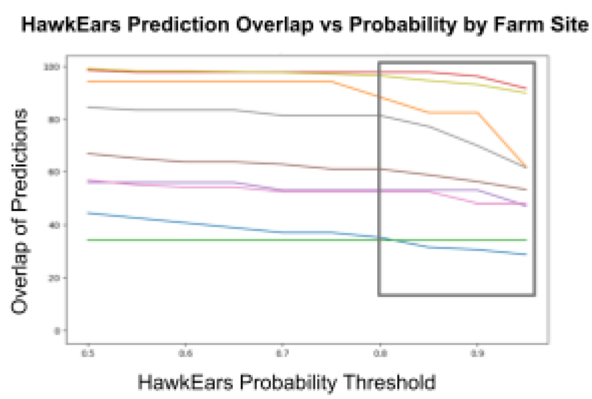
THE UNIVERSITY OF CHICAGO
DATA SCIENCE INSTITUTE

Ecdysis Foundation is a non-profit focused on regenerative agriculture, including farming practices that boost biodiversity. Ecdysis examines biodiversity through bird species identification. They are interested in automating species identification using audio recordings collected from over 1000 farms.

Our team applied four audio classification models to the audio files. These models predicted: 1) bird species in the audio, 2) where in the audio they were identified, 3) certainty probability of the detection. We compared the predictions, accuracy, and precision across all models. We also compared prediction results to the visually collected bird observation data provided by Ecdysis. Our best performing models are visualized below:




HawkEars detected a significantly higher number of species compared to BirdNET. This is plotted alongside the count of detected birds by visual data.




Across 9 different farm sites, BirdNET's bird detections were largely contained within a subset of HawkEars's strongest bird predictions.

This initial analysis clarifies how three different bird detection models effectively detect bird species. There is an overlap in predictions between both the models. However, one model captures a greater share of visual bird observations. Our findings provide insight into a more efficient way in which Ecdysis can autonomously identify bird species.